Justify all answers. Submit as a single PDF.

1. What is the probability that a five-card poker hand drawn from a standard deck
   (a) (5 points) contains two pairs (that is, two of each of two different kinds and a fifth card of a third kind)?
   (b) (5 points) contains a flush, that is, five cards of the same suit?
   (c) (5 points) contains a straight, that is, five cards that have consecutive kinds? (Note that an ace can be considered either the lowest card of an A-2-3-4-5 straight or the highest card of a 10-J-Q-K-A straight.)
   (d) (5 points) contains a straight flush, that is, five cards of the same suit of consecutive kinds?
   (e) (5 points) contains a royal flush, that is, the 10, jack, queen, king, and ace of one suit?

2. In roulette, a wheel with 38 numbers is spun. Of these, 18 are red, and 18 are black. The other two numbers, which are neither black nor red, are 0 and 00. The probability that when the wheel is spun it lands on any particular number is $\frac{1}{38}$.
   (a) (5 points) What is the probability that the wheel lands on a red number?
   (b) (5 points) What is the probability that the wheel lands on a black number twice in a row?
   (c) (5 points) What is the probability that the wheel lands on 0 or 00?
   (d) (5 points) What is the probability that in five spins the wheel never lands on either 0 or 00?
   (e) (5 points) What is the probability that the wheel lands on one of the first six integers on one spin, but does not land on any of them on the next spin?

3. What is the probability of these events when we randomly select a permutation of $\{1, 2, 3, 4\}$?
   (a) (5 points) 1 precedes 4.
   (b) (5 points) 4 precedes 1.
   (c) (5 points) 4 precedes 1 and 4 precedes 2.
   (d) (5 points) 4 precedes 1, 4 precedes 2, and 4 precedes 3.
   (e) (5 points) 4 precedes 3 and 2 precedes 1.

4. What is the conditional probability that exactly four heads appear when a fair coin is flipped five times,
   (a) (5 points) given that the first flip came up heads?
   (b) (5 points) given that the first flip came up tails?

5. Let E and F be the events that a family of n children has children of both sexes and has at most one boy, respectively. Are E and F independent if

   (a) (5 points) $n = 2$?

   (b) (5 points) $n = 4$?

   (c) (5 points) $n = 5$?

6. (10 points) Show that if E and F are events, then $Pr[E \cap F] \geq Pr[E] + Pr[F] - 1$. This is known as Bonferroni's inequality.

7. (10 points) Use mathematical induction to prove the following generalization of Bonferroni's inequality:

$$Pr[E_1 \cap E_2 \cap \cdots \cap E_n] \;\geq\; Pr[E_1] + Pr[E_2] + \cdots + Pr[E_n] - (n-1),$$

   where $E_1, E_2, \ldots, E_n$ are n events.